

Yufan Du

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Education

Peking University **Sep. 2021 - Present**
Bachelor of Science in Applied Physics *and* Computer Science Dual Major *Beijing, China*

Relevant Scores

Overall GPA: 3.844/4.0 (top 10% of 66 students)
TOEFL: 108(R27 + L30 + S23 + W28)
GRE: 324 (Quantitative + Verbal), 4.0 (Analytical Writing)

Technical Skills

Knowledgeable with: C/C++, Python
Familiar with: Python ML libraries (PyTorch, NumPy), Verilog, Embedded systems, FPGA, SQL/Database, JavaScript/HTML, Linux systems, MATLAB, Tcl, $\text{\LaTeX}$
EDA tools: Familiar with Cadence Innovus, Cadence Virtuoso

Project Experiences

FPGA-Based Snake Game Design Project **Mar. 2023 - Jun. 2023**

- Advisor: Professor Xiaohui Duan.
- Deployed a computer vision algorithm and the classic Snake game on a resource-limited FPGA platform, optimizing resource usage.
- Developed and implemented a real-time control mechanism for the game using hand movement detection with a camera, enhancing user interaction.
- Introduced a 3D human-computer interface to improve player engagement and provided valuable insights for IoT edge device development.

AI-Assisted Schedule Manager **May. 2023 - July. 2023**

- Advisor: Professor Wei Guo.
- Developed a cross-platform scheduling application integrated with AI, applying NLP for user-friendly schedule management with commercial API.
- Achieved a substantial improvement in user experience by enabling efficient management and reminders through conversational interfaces.

Research Experiences

Cross-Stage Power Prediction for Integrated Circuits **Jun. 2023. - Nov. 2023.**

- Advisor: Professor Yibo Lin.
- Served as the first author.
- Proposed an innovative VLSI circuit power prediction framework that integrates cross-stage circuit-transformation-aware learning.
- Achieved a significant reduction in error rates and computation time compared to industry-leading commercial tools.
- It thus enabled effective power optimization strategies in early-stage VLSI design.
- *Accepted by DAC 2024.*

Publications

- **Yufan Du**, Zizheng Guo, Xun Jiang, Zhuomin Chai, Yuxiang Zhao, Yibo Lin, Runsheng Wang and Ru Huang (2024). PowPrediCT: Cross-Stage Power Prediction with Circuit-Transformation-Aware Learning. *2024 Design Automation Conference (DAC24)*.